Prof-interest

A Project on Retrieving and Compiling Professors' Information

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1 Introduction

College serves as the opportunity for many to develop their expertise towards a future career, and this growth occurs through both coursework and professor-led research. However, discovering the right research opportunities is quite the challenge. Even at a smaller institution like Pomona College, a faculty size of 286 is simply overlooked by many students who do not even know where to start searching, on top of their busy schedules. Prof-interest aims to assist students in connecting with faculty who share their specific research interest. Using Prof-interest, professors searchable by research topics, sorted by identifying information, and mapped across interdisciplinary fields are just a few clicks away.

Our Github repository is available at https://github.com/soppark/stalkMyProfessors

2 Background

A brief brainstorming session yielded the topic of student-faculty interaction as the project's focus. In accordance with the human centered design paradigm, needfinding interviews were conducted with three students across disciplines at Pomona College: a Chemistry major, a Math major, and a Music and Media Studies double major.

Each interviewee was asked a series of questions such as "How do you learn about professors?" and "Describe a time when you were interested in a professor's research." When the results were compiled, a majority of the interviewees shared stories of discovering perfect match research opportunities by chance. For example, one student heard their professor talk about their research in class one day and went to office hours to learn more. Another was just chatting with their professor when they suddenly offered them a summer research opportunity that they thought they were suited for. These discoveries revealed that there are many interesting research opportunities, but students have difficulty discovering them.

Digging deeper, it was found that professors' research interests and papers are stored across a series of difficult mediums: school websites, personal websites, scholarly databases, and more. It was clear that the current method to discover the best professor research was cumbersome. This led to the creation of the project's guiding design question: "how might we easily connect students with professors who share their interests?"

Prof-interest emerged from this empathizing process as a system that would serve

as a one stop location for students to search and sort professors based on their interests in order to efficiently identify the best professor for them to talk with. After consulting with Computer Science professors Li and Clark, a third feature to visualize which professors and disciplines are more connected with other fields or not on the research side was added to the project.

3 Data and Structures

Each item in the dataset is a research paper written by a Pomona College faculty member. In addition to the professor and paper's names, each data point also includes contact information, the professor's title, publishing venue, research interests, and the names of the paper's coauthors at Pomona College.

The data was retrieved from Google Scholar using the Python scholarly library. A parsing Python function wrote all information to a txt file, with a new txt file created for each professor. Next, a second Python function combined each file into a master CSV file that includes all of the data points. A series of coauthors were added manually to the CSV file. Coauthor information was generated via a large language model and appended to the master CSV file (see section on citations and AI usage).

A CSV format was chosen because it was decided to be the easiest to run through with a script, and because the project group members had the most experience working with CSV files compared to other types of data storage.

Data for this project is stored primarily in three data structures: Hashtables, Binary Search Trees, and Graphs.

Hashtables store information on professor names and their `Prof` object with more details.

Binary Search Trees are used for showing professors within one department, specifically the sorting feature by number of papers found.

Graphs are used solely for representing coauthors as the edges can very cleanly describe co-author relationships with easy visualization. We used built-in `javax.swing` and `java.awt` to make simple graphs. Swing creates the window and awt allows us to draw circles, lines, and professor names.

4 Results

Below, we have the screenshots for the 3 features we have.

4.1 Feature 1:

The first feature is the easiest, retrieving one professor's details, including their department, past paper title and other information like email address.

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Enter 1 to search Prof, 2 for Dept, 3 for whole graph, 4 for one prof graph, 0 for prof list: 1
Enter the name of the Professor you'd like to learn more about: Anne Dwyer
Name: Anne Dwyer
Affiliation: "Associate Professor of Russian, Pomona College"
Department: Russian
Email: @pomona.edu
Website:
Academic Interestst:
Comparative Literature
Russian Literature
German Literature
Cultural History
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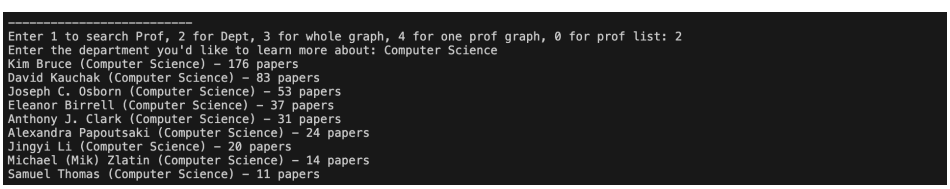
18 Past Papers Found:
Title: Standstill as Extinction: Viktor Shklovsky's Poetics and Politics of Movement in the 1920s and 1930s, Year: 2016, Venue: "PMLA 131 (2), 269–288, 2016"
Title: "Revivifying Russia: Literature, Theory, and Empire in Viktor Shklovskii's Civil War Writings", Year: 2009, Venue: "slavonica 15 (1), 11–31, 2009"
Title: Dostoevsky's Prison House of Nation(s): Genre Violence in Notes from the House of the Dead, Year: 2012, Venue: "The Russian Review 71 (2), 209–225, 2012"
Title: Runaway Texts: The Many Life Stories of Iurii Trifonov and Christa Wolf, Year: 2005, Venue: "The Russian Review 64 (4), 605–627, 2005"
Title: Of Hats and Trains: Cultural Traffic in Leskov's and Dostoevski's Westward Journeys, Year: 2011, Venue: "Slavic Review 70 (1), 67–93, 2011"
Title: "The Multilingual Pleasures of Slavic Worlds: Sacher-Masoch, Franzos, Freud", Year: 2013, Venue: "Comparative Literature 65 (2), 137–161, 2013"
Title: "Improvising empire: Literary accounts from the Russian and Austrian borderlands, 1862–1923", Year: 2007, Venue: "University of California, Berkeley, 2007"
Title: "The Origins of Russian Literary Theory: Folklore, Philology, Form. By Jessica Merrill. Evanston: Northwestern University Press, 2022. ix, 312 pp. Appendix. Notes. Bibliograp
hy …", Year: 2023, Venue: "Slavic Review 82 (2), 564–565, 2023"
Title: "Permanent Evolution. Selected Essays on Literature, Theory and Film", Year: 2021, Venue: "The Slavic and East European Journal 65 (1), 196–197, 2021"
Title: "Dialogues with Shklovsky: The Duvakin Interviews, 1967–1968", Year: 2019, Venue: "The Russian Review 78 (4), 661–662, 2019"
Title: "Optical Play: Glass, Vision, and Spectacle in Russian Culture. Studies in Russian Literature and Theory", Year: 2017, Venue: "The Russian Review 76 (2), 352–353, 2017"
Title: "From the Shadow of Empire: Defining the Russian Nation through Cultural Mythology, 1855–1870. Studies of the Harriman Institute", Year: 2011, Venue: "The Slavic and East Eur
opean Journal 55 (4), 647–649, 2011"
Title: FEATURED REVIEW 165 FILM REVIEWS 168, Year: 2011, Venue: "Slavic Review 70 (3), 2011"
Title: "Orientations: An Anthology of East European Travel Writing, ca. 1550–2000. East Looks West, Vol. 1", Year: 2010, Venue: "The Slavic and East European Journal 54 (4), 711–712
, 2010"
Title: "Under Eastern Eyes: A Comparative Introduction to East European Travel Writing on Europe. Series: East Looks West, vol. 2", Year: 2009, Venue: "The Slavic and East European
Journal 53 (4), 719–721, 2009"
Title: "Wendy Bracewell and Alex Drace-Francis, eds. Under Eastern Eyes: A Comparative Introduction to East European Travel Writing on Europe", Year: 2009, Venue: "Slavic and East E
uropean Journal 53 (4), 719, 2009"
Title: Editor: Disturbed Aesthetic Presence: In Avantgarde and Late Avantgarde, Year: 2008, Venue: "SLAVIC AND EAST EUROPEAN JOURNAL 52 (4), 639–641, 2008"
Title: "fRaktur: Gestörte ästhetische Präsenz in Avantgarde und Spätavantgarde. Wiener Slawistischer Almanach, Literarische Reihe, Special Issue 63. Intermedia Series, Issue 3", Yea
r: 2008, Venue: "The Slavic and East European Journal 52 (4), 639–641, 2008"
```

Figure 1: Feature 1 Screenshot

4.2 Feature 2:

For the second feature, we display the professors within one specific department in descending order of past papers. This feature attempts to help people find their advisor or professors to reach out for a specific academic field of interest.



```
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Enter 1 to search Prof, 2 for Dept, 3 for whole graph, 4 for one prof graph, 0 for prof list: 2
Enter the department you'd like to learn more about: Computer Science
Kim Bruce (Computer Science) - 176 papers
David Kauchak (Computer Science) - 83 papers
Joseph C. Osborn (Computer Science) - 53 papers
Eleanor Birrell (Computer Science) - 37 papers
Anthony J. Clark (Computer Science) - 31 papers
Alexandra Papoutsaki (Computer Science) - 24 papers
Jingyi Li (Computer Science) - 20 papers
Michael (Mik) Zlatin (Computer Science) - 14 papers
Samuel Thomas (Computer Science) - 11 papers
```

Figure 2: Feature 2 Screenshot

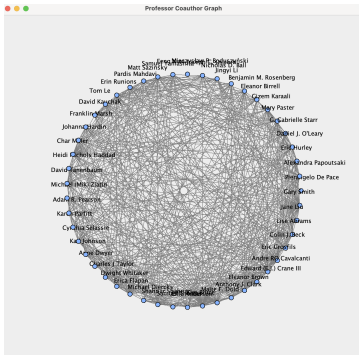
4.3 Feature 3:

the graph represents the coauthor network between professors. Each vertex in the graph is one professor from our dataset. Each edge represents a coauthor relationship.

4.3.1 Option 3 (Whole coauthor graph)

For the whole graph, we include all professors in the dataset and connect professors who are listed as coauthors. For the one-professor graph, we include the selected professor, their coauthors, and any connections between these coauthors.

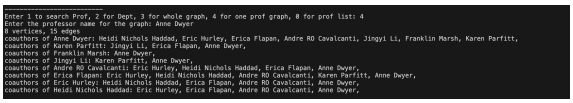
(a) Feature 3



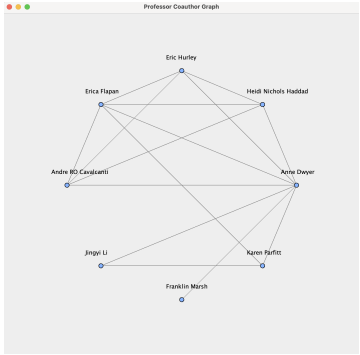
(b) Feature 3 Visualize

Figure 3: Feature 3 part one: All Prof Graph

4.3.2 Option 4 (One prof's graph)



(a) Feature 3



(b) Feature 3 Visualize

Figure 4: Feature 3 part two: One Prof Graph

5 Performance Analysis

Loading Professor Data: For all features, the basic pre-condition is that the CSV file has to be read correctly. The program basically depends on `ProfList` to build the `HashMap<String, prof>` and each prof name points to one `Prof` object. If the file is missing, formatted badly, or prof names are inconsistent, the rest of the features will not work well.

The time complexity is about $O(n)$ because the program reads each CSV row and uses `HashMap` to group rows by professor name. The space complexity is also $O(n)$, since the program stores all professors.

Listing All professors: Before this feature runs, the professor map has to already be created from the CSV file.

The time complexity is $O(n)$ because it loops through every professor once and prints their basic information. The space complexity if $O(1)$ extra space, since it does not create a new major data structure, and it only uses the existing `HashMap`.

Searching For One Professor: Before feature 1 runs, the user has to type the professor's name exactly as it appears in the dataset. If the name is not found, the program prints an error message.

The time complexity is $O(1)$ on average for the `HashMap` searching. Printing the professor details takes $O(n)$ because it prints all of the professor's papers. The space complexity is $O(1)$ extra space because the program only stores the user input and the existing data.

Searching By Department: Before this runs, the professor data has to be loaded, and the user has to enter a department name.

The time complexity for scanning all professors is $O(n)$, and for inserting the matching professors into the BST and removing them in sorted order is $O(\log n)$, if the BST is balanced. In worst case, it can be $O(n^2)$, because the BST is not self-balanced. The space complexity is $O(n)$, because we store the professors that belong to the department.

Building One Professor Graph The professor name must be typed exactly right.

The time complexity is about $O(n{\cdot}n)$ because it depends on the size of the professor's network and the program checks coauthors and `addEdge` between coauthors. The space complexity is $O(V + E)$, because it only stores the professor, their coauthors and the edges between them.

Building Whole coauthor Graph Before this feature runs, the professor ap must be loaded and the coauthor data must exist in the CSV file.

The time complexity is $O(n \cdot n)$, which is similar to building one professor's graph, because it loops through every professors and coauthors. The space complexity is also $O(V + E)$ because the graph stores vertices and edges.

6 Affordance Analysis

The primary benefits of this project are the students who will be able to search their professors' research easier thanks to the Prof-interest/StalkMyProfessor's search, sort, and visualization tools. The knock-on effect of this project is the enhanced visibility it affords professors, who will be able to access a larger pool of students to support their research projects. Most college students poses personal devices that can run this software, but those without access to a device with a JVM will be left behind by this project.

By using the metric of papers published for sorting, this project reinforces the conception that those who publish more often are the better professors to work with, when this is not always true. Additionally, the software has trouble accounting for professors who interact with multiple fields beyond their primary expertise, as searches only identify professors by their listed position/main research interests.

A malicious use of this software could be using its provided information to feign interest in a professor's work to get a research position for ulterior motives such as SURP money or lab access. It could also be utilized as a tool to stalk college faculty by giving perpetrators quick access to the interests, field of work, and contact information of professors. However, this threat should be mitigated by safeguarding the Prof-interest/StalkMyProfessor software.

7 Citations and AI Usage

Some structures are edited based on existing java classes from Princeton Algorithms course, retrived from: Graph.java, BST.java, and Bag.java. We edited from the basic structures and methods useful in our program, and removed some irrelevant methods for brevity.

For data processing, we used ChatGPT to generate a Python script named `parse.py`. This compiled the separate `.txt` files we had for each individual professors into one single `.csv` file that is easier for reading and processing.

Another usage of Generative AI is to create another Python script to generate coauthors of each paper since we didn't find an easy way to scrap this data with API. Another concern was also if professors had coauthors outside of the professors we have in data. Then, a problem arises of how to represent these outside professors. Hence, for simplicity we prompted Codex to generate a script that can randomly create 0-2 coauthors within the professor names that we have in database. This generated Python file is called `add_coauthors.py`.

The two prompts that generated these two files can be found in github and is available at [1]. A conversation link isn't available but the session ID is provided.

8 Reflections

This project took longer than expected due to certain difficulties with curating the data. However, we had correctly predicted that the data collection process would take much longer than the coding process. The project took 35 days: spanning from the first brainstorming lab to the due date on May 13. Prof-interest/StalkMyProfessor stuck fairly close to its originally proposed idea. However, some features achieved less functionality than the original stretch goals of the project. For example, professor sorting only works by the number of papers published.

The project still employs the same data structures that were discussed at the pitch meeting with Professor Li. We were unable to implement the extra credit task of a search and sort GUI in time, but all others tasks from the grading contract were successfully fulfilled. Overall, this project was a positive experience that yielded a program that–with minor tweaks–could serve as a proof of concept for a product that would help many students and professors find their best matches for in-school and summer research experiences.

References

[1] Max Liu et al. *cs62-Final-Project-Data-Generation*. May 2026. URL: https://github.com/Wanponfer/cs62-Final-Project-Data-Generation.